

GeNRe: A French Gender-Neutral Rewriting System Using Collective Nouns

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1. Introduction

- Masculine generics (MG) in gender-marked languages (e.g., French, German, Dutch): use of the masculine form as a default/neutral form to refer to mixed group of men/women. Examples in French:

(a) « Les **étudiants**^{MG} sont partis. » (**Students**^{MG} left.)

- Psycholinguistics studies show that **MG induce gender bias** and **amplify male-centric mental representations**^[1,2]
- Task of **gender rewriting**: propose alternatives to gendered sentences to balance datasets and reduce use of MG and related biases

2. Task of Gender Rewriting

- Three types** of existing gender rewriting systems: **neutral** rewriting (English^[3,4]), **inclusive** rewriting (French^[5], Portuguese^[6], German^[7]) and **gender-switching** rewriting (Arabic^[8])
- No neutral rewriting system for French**, and **collective nouns (gender-fixed in French) not tested** as potentially good candidates for gender-neutralization
- Automatically switching from MG to collective nouns in French requires **number and gender changes, re-inflection**

Masculine generics

Tous les anciens employés^{MG}
sont invités au programme.
(All former employees^{MG}
are invited to the program.)

Collective noun

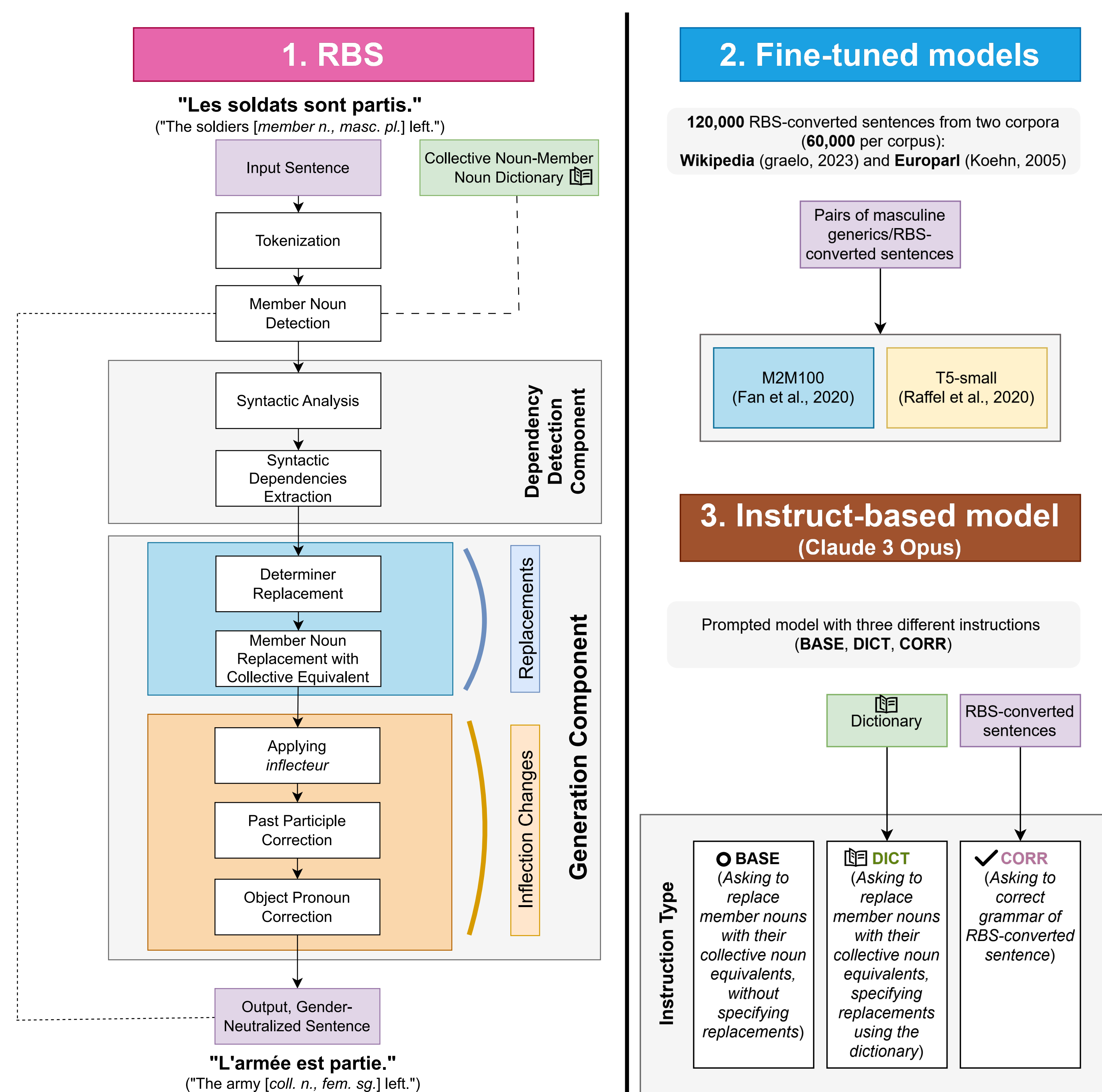
L'ancien personnel
est invité au programme.
(Former staff
is invited to the program.)

3. Methodology

- Create a French member noun-collective noun dictionary based on the work of **Lecolle (2019)**^[9], corpus search and scraping (**315 entries** total)

Member noun	Collective noun
employés (<i>employees</i>)	personnel (<i>staff</i>)
jurés (<i>jury members</i>)	jury
policiers (<i>police officers</i>)	police
soldats (<i>soldiers</i>)	armée (<i>army</i>)
...	...

- Use 3 approaches: RBS (*spaCy*^[10] for syntactic dependency detection; *inflecteur*^[11] for re-inflecting); fine-tuned models (M2M100/T5) and instruct-based model (Claude 3 Opus)



4. Evaluation and Results

- Evaluation dataset of 500 sentences (250 Wikipedia/250 Europarl)
- RBS Dependency Detection Component**: Manually annotated syntactic dependencies in evaluation sentences
 - Objective: evaluate correct detection of syntactic dependencies of the member noun to be modified
 - Baseline: default spaCy-detected member noun's dependencies (excluding punctuation)

	Wikipedia			Europarl			Avg.		
	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1
Baseline	0.096	0.723	0.169	0.115	0.689	0.197	0.1055	0.706	0.183
GeNRe-RBS	0.773	0.855	0.812	0.758	0.813	0.785	0.7655	0.834	0.7985

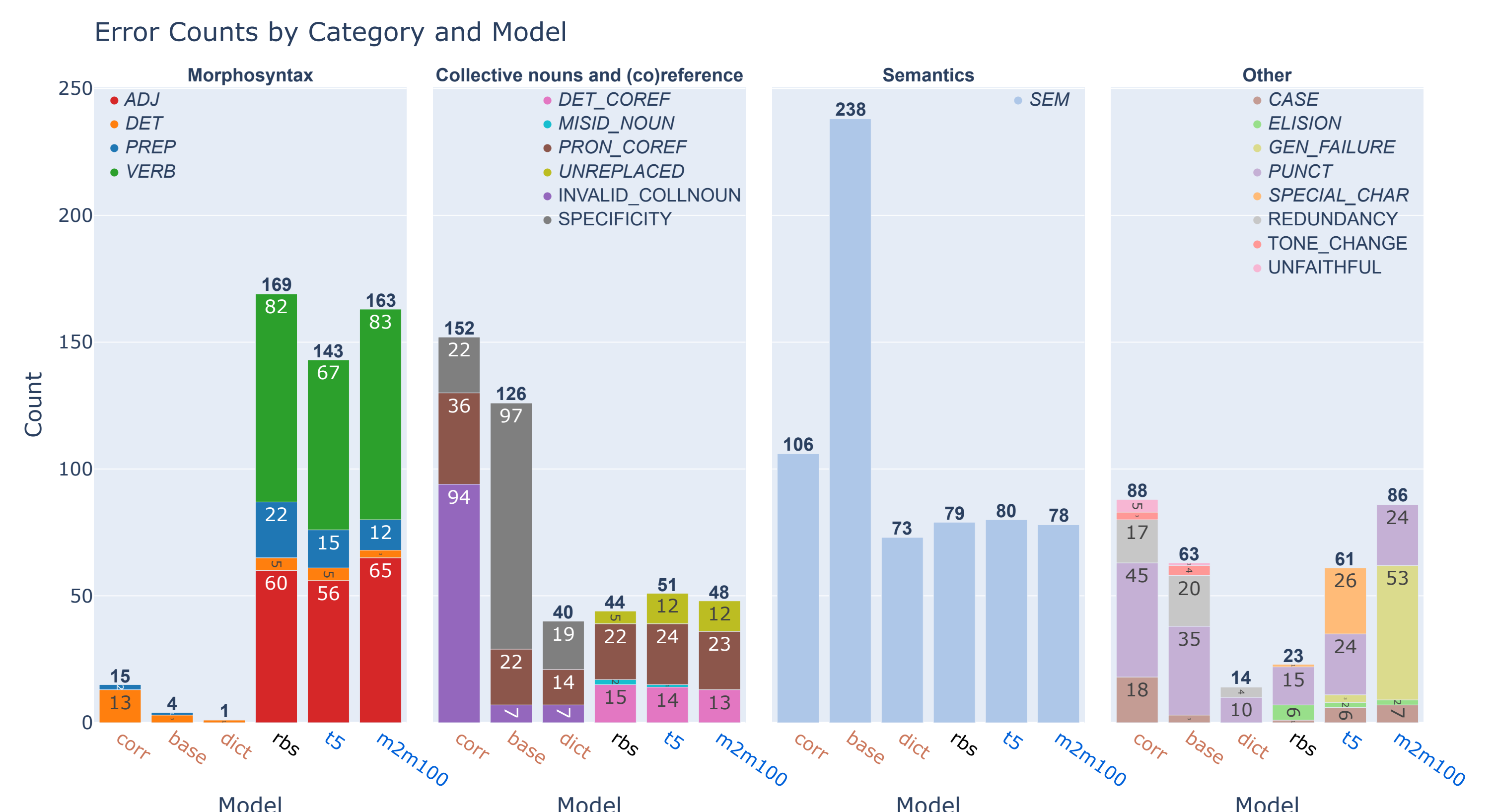
RBS Dependency Detection Component Results

- Generation (All Models)**: Manually gender-neutralized evaluation sentences using collective nouns
 - Objective: evaluate correct conversion of sentences
 - Baseline: original, unconverted sentence (as per previous works)

	Wikipedia			Europarl			Avg.		
	WER (↓)	BLEU (↑)	Cosine (↑)	WER (↓)	BLEU (↑)	Cosine (↑)	WER (↓)	BLEU (↑)	Cosine (↑)
Baseline	12.611%	80.688	97.436	12.446%	82.871	97.008	12.529%	81.779	97.222
GeNRe-RBS	4.03%	92.096	98.88	3.814%	93.707	99.22	3.81%	92.887	99.05
GeNRe-T5	6.726%	87.358	98.508	4.259%	93.111	99.1	5.492%	90.234	98.804
GeNRe-M2M100	6.566%	88.186	97.232	4.247%	93.197	98.992	5.406%	90.692	98.112
Claude-BASE	13.87%	80.205	96.532	10.713%	85.313	97.128	12.291%	82.759	96.83
Claude-DICT	4.702%	92.79	98.812	4.197%	94.247	99.264	4.45%	93.519	99.038
Claude-CORR	11.282%	84.954	98.092	8.992%	85.257	98.056	10.137%	85.25	98.074

Global Generation Results

- Generation Errors**: For RBS and fine-tuned models, two annotators manually annotated with subcategories (italicized in plot)
- For Claude 3 Opus, used LLM-as-a-judge to automatically annotate in a two-step process: 1. ask to describe the error; 2. assign a subcategory



5. Takeaways

- RBS and instruct-based model combined with pre-created resources achieve similar results; lower morphosyntactic errors for Claude 3 Opus
- Collective nouns useful for gender neutralization, but limited by restrictive semantics

References

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